Operating System

LAB 1

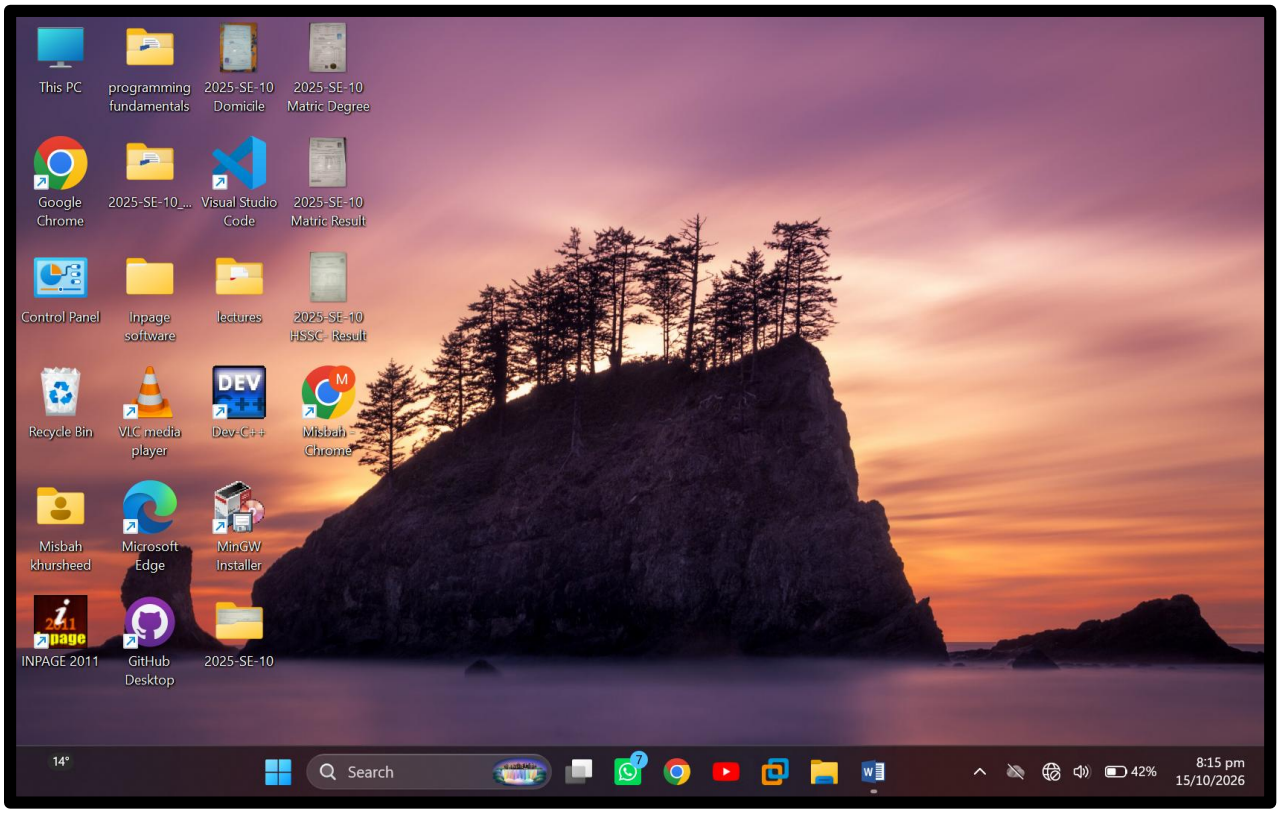
Introduction:

OBJECTIVES:

1. Operating System Introduction
2. Cores
3. 32 bit and 64 bit operating systems
4. Windows operating systems
5. Task manager
6. Operating systems services

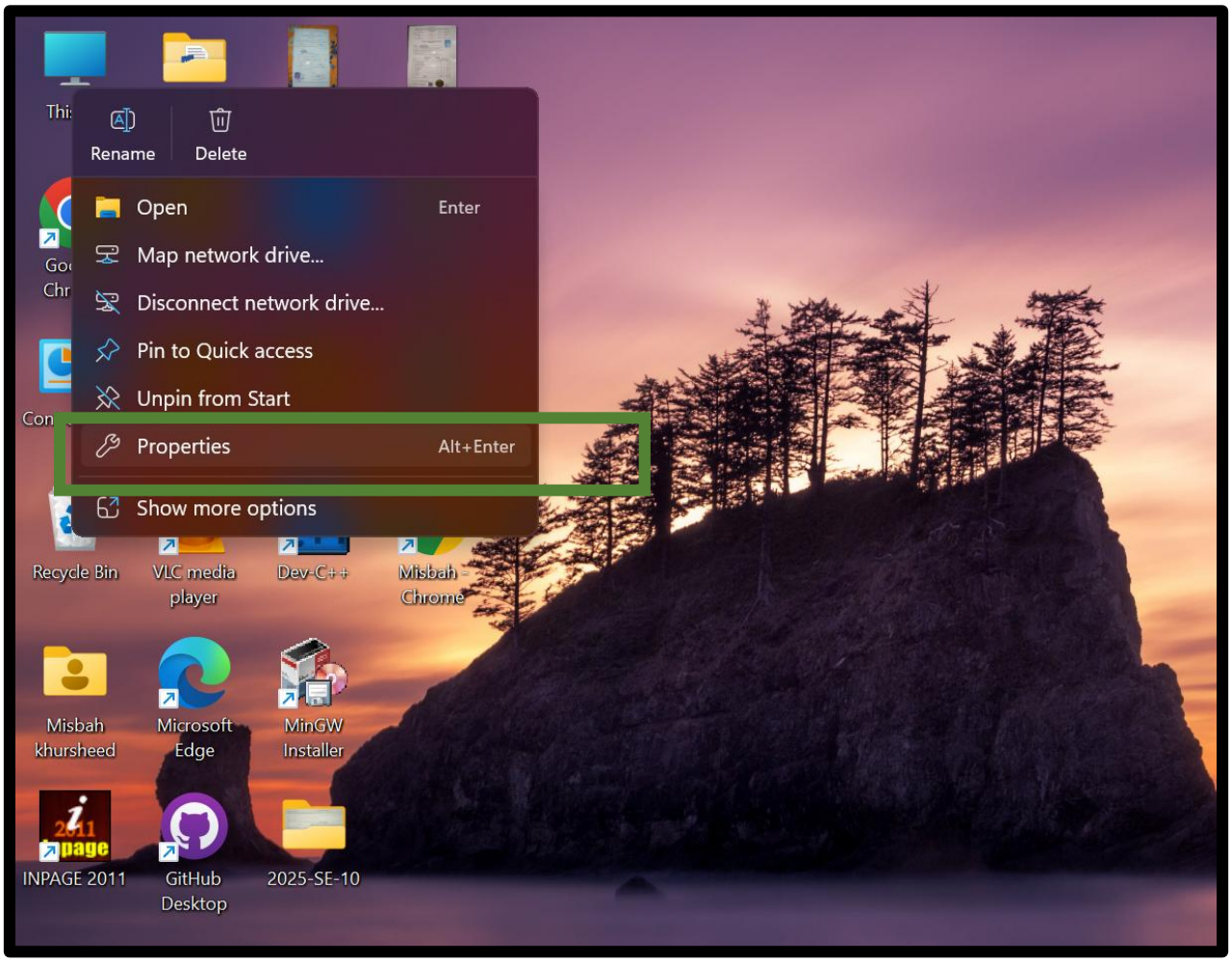
Operating systems installed in computer:

Windows 11

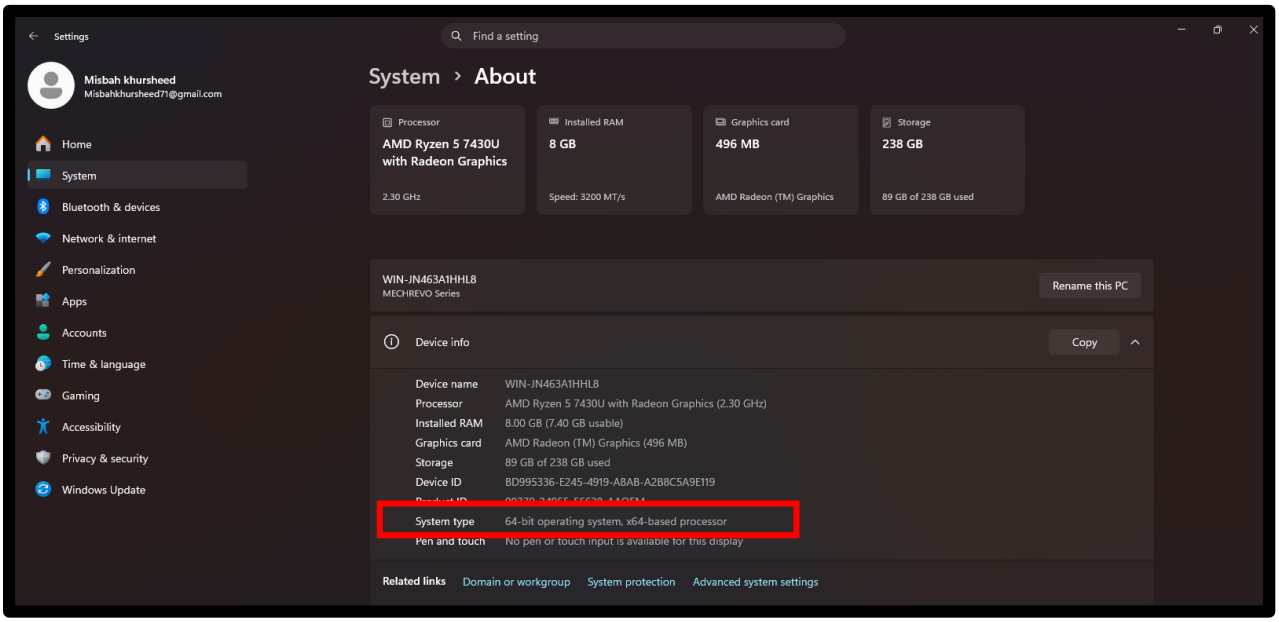


Check the version of Operating System (In windows):

- Click on Computer icon. And select the properties



For a 64-bit version operating system, 64-bit Operating System appears for the System type under System. For a 32-bit version operating system, 32-bit Operating System appears for the System type under System.

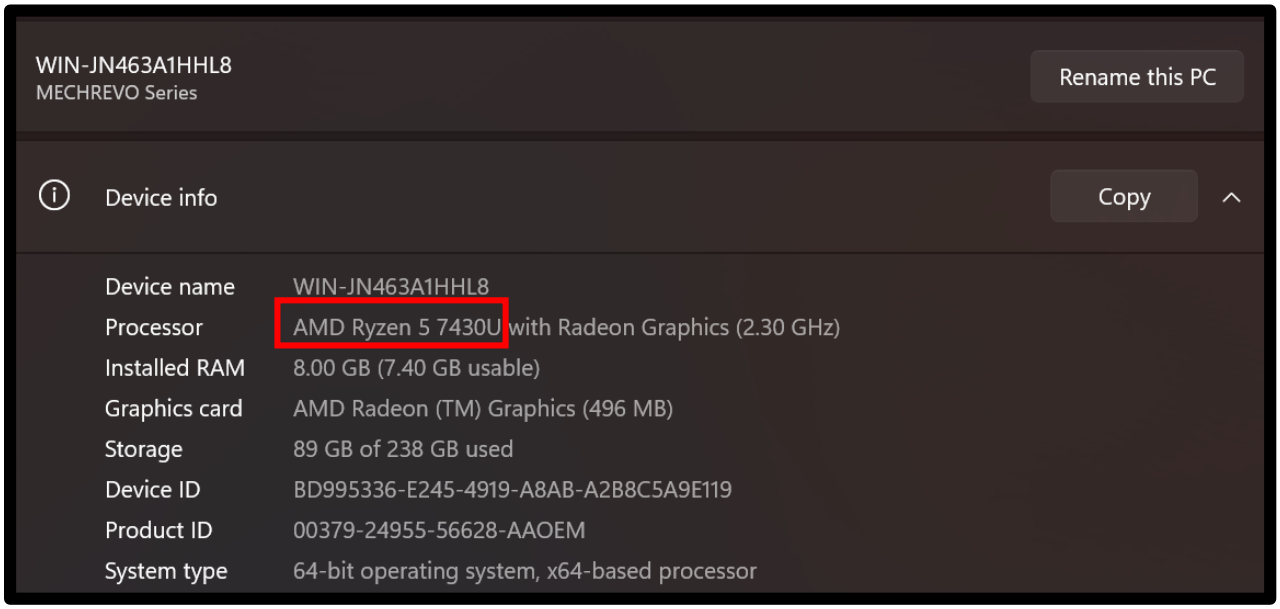


Cores of Processor:

A core, or CPU core, is the "brain" of a CPU. It receives instructions, and performs calculations, or operations, to satisfy those instructions. A CPU can have multiple cores. A processor with two cores is called a dual-core processor; with four cores, a quad-core; six cores, hexa-core; eight cores, octa-core.

A core is a small CPU or processor built into a big CPU or CPU socket. It can independently perform or process all computational tasks. From this perspective, we can consider a core to be a smaller CPU or a smaller processor within a big processor.

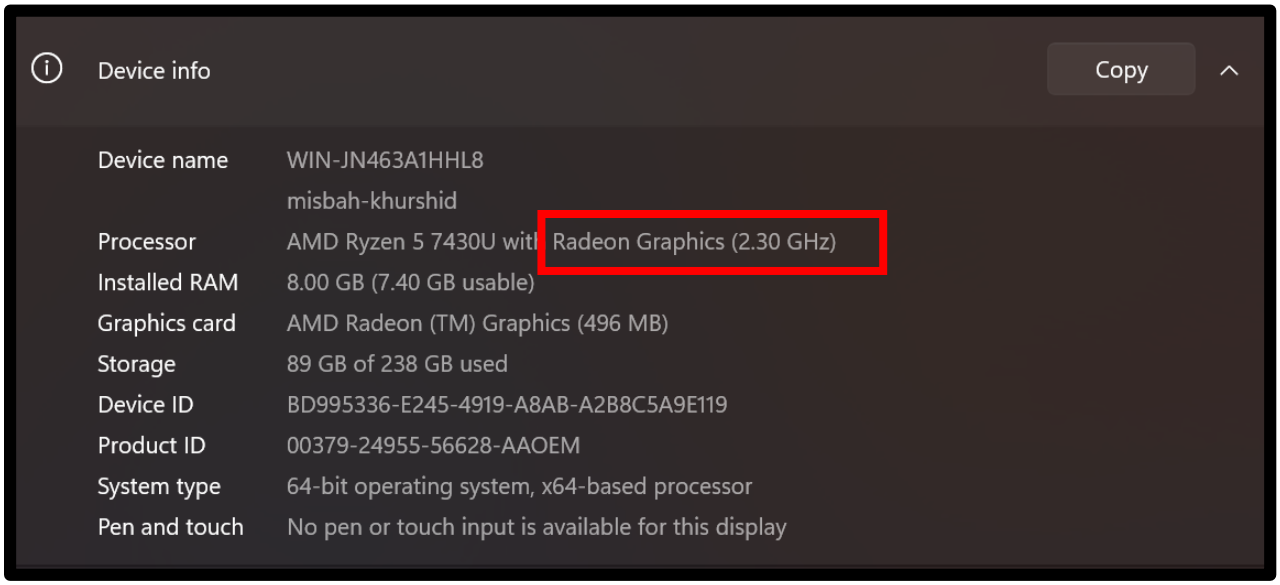
The main **difference between CPU** and **Core** is that the **CPU** is an electronic circuit inside the computer that carries out instruction to perform arithmetic, logical, control and input/output operations while the **core** is an execution unit inside the **CPU** that receives and executes instruction.



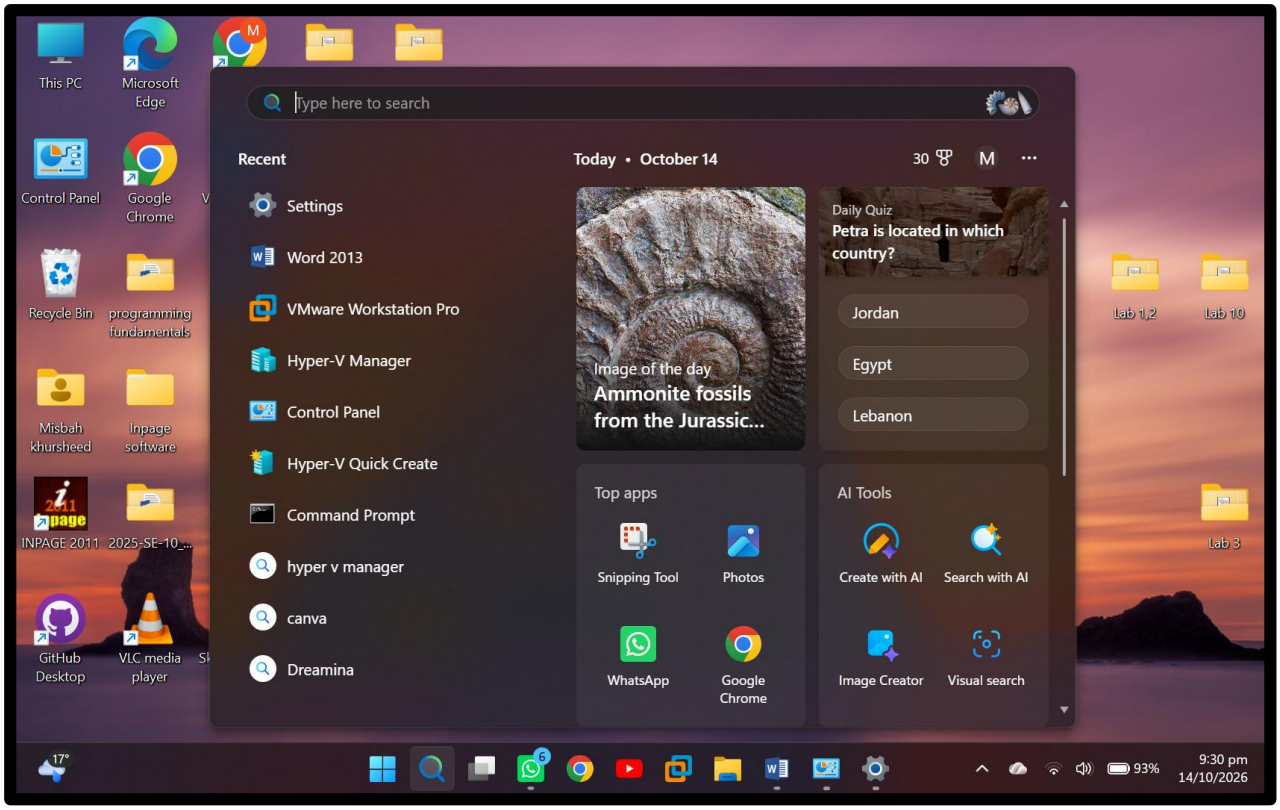
Processor Clock Speed:

A computer's **processor** clock **speed** determines how quickly the central **processing** unit (**CPU**) can retrieve and interpret instructions. This helps your computer complete more tasks by getting them done faster. Clock speeds are measured in gigahertz (GHz), with a higher number equating to higher clock **speed.**

The clock speed of my laptop's processor is shown in red box shown in figure below.



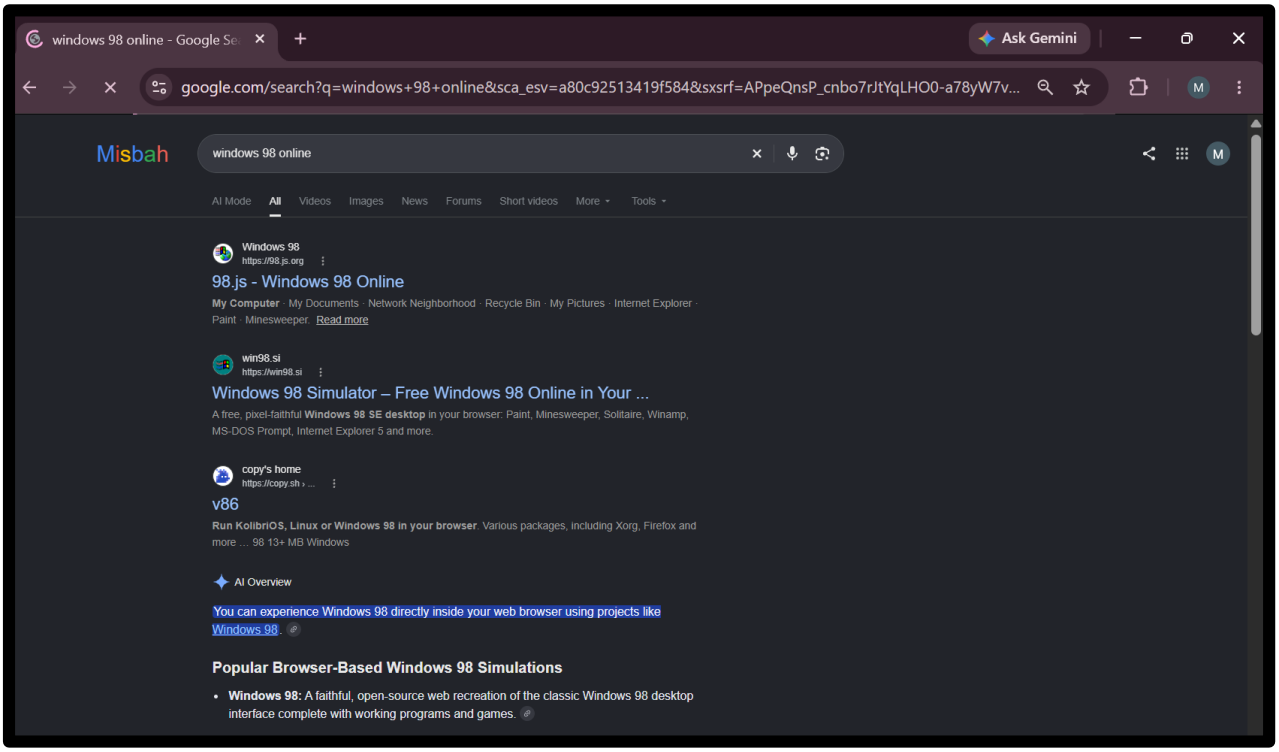
Windows Operating System:



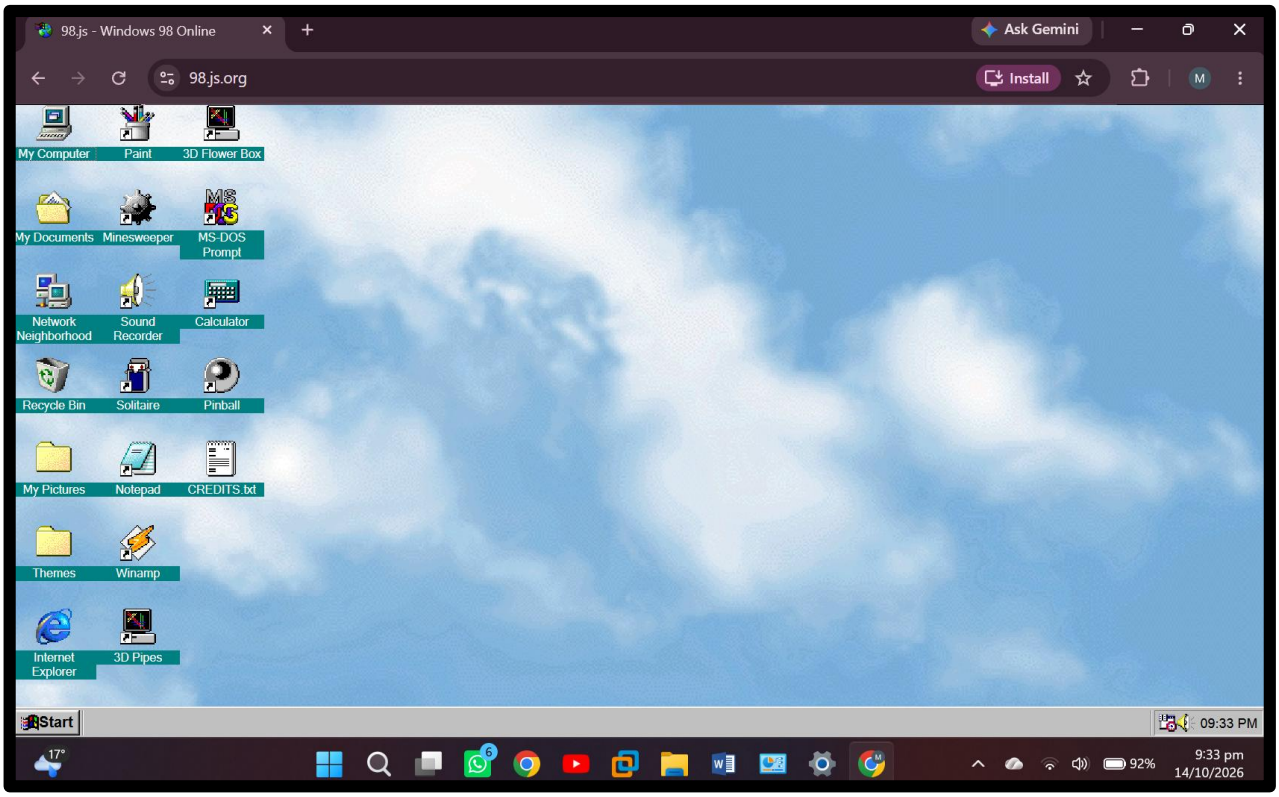
Online Operating Systems:

Windows 98:

To see the interface of Windows 98, go online and write Windows 98 online

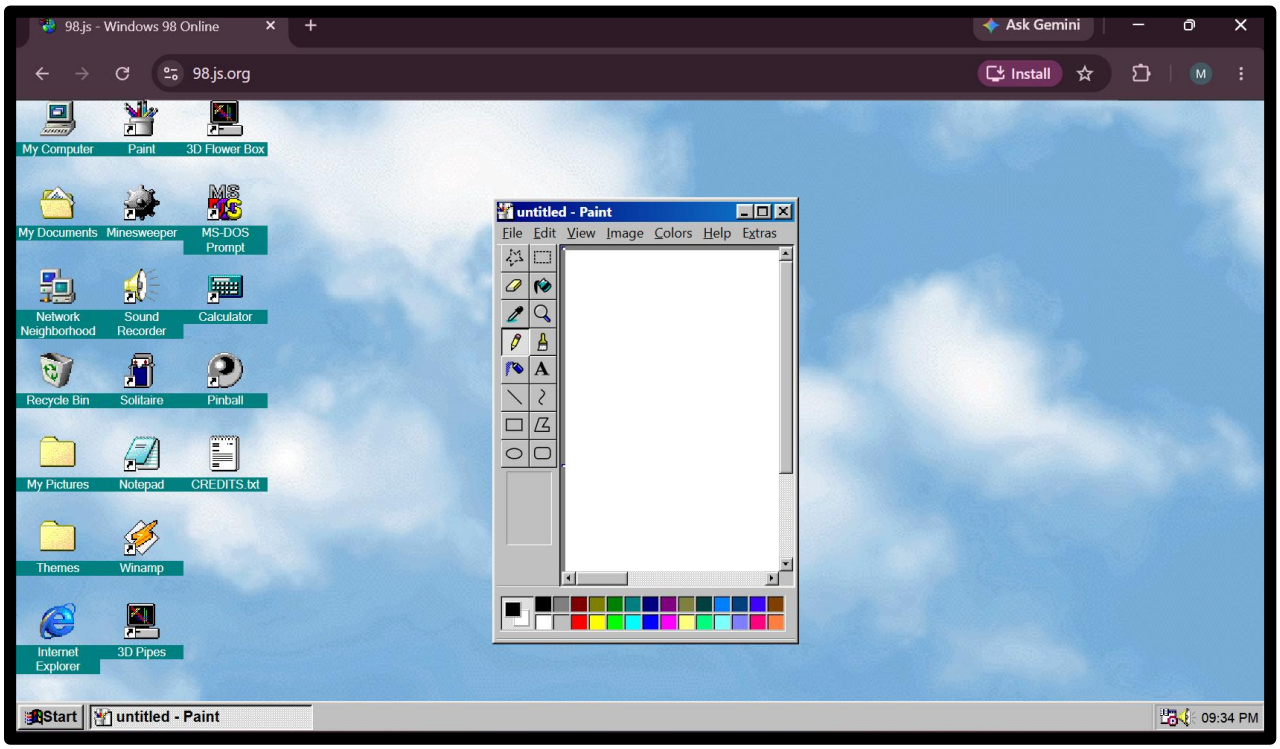


Click on Windows online 98.js



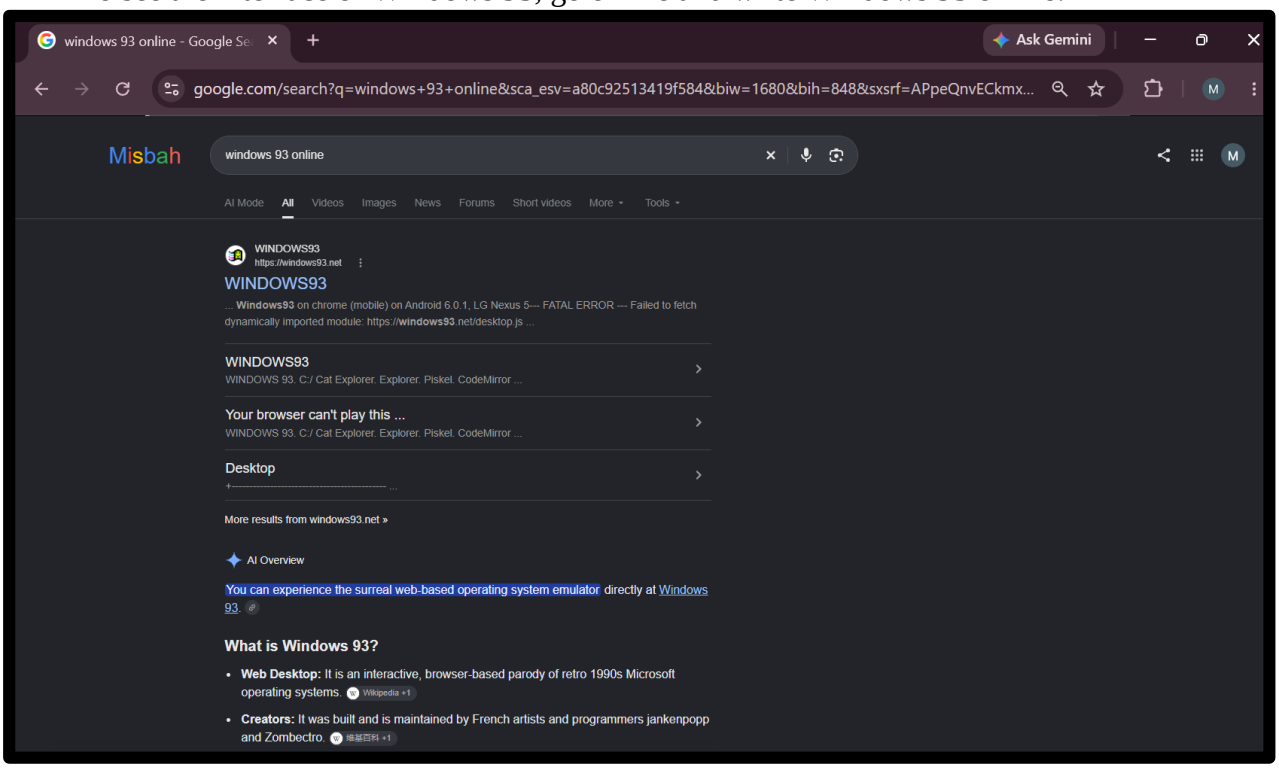
You can click on icons and see how windows 98 was working.

Click on Paint

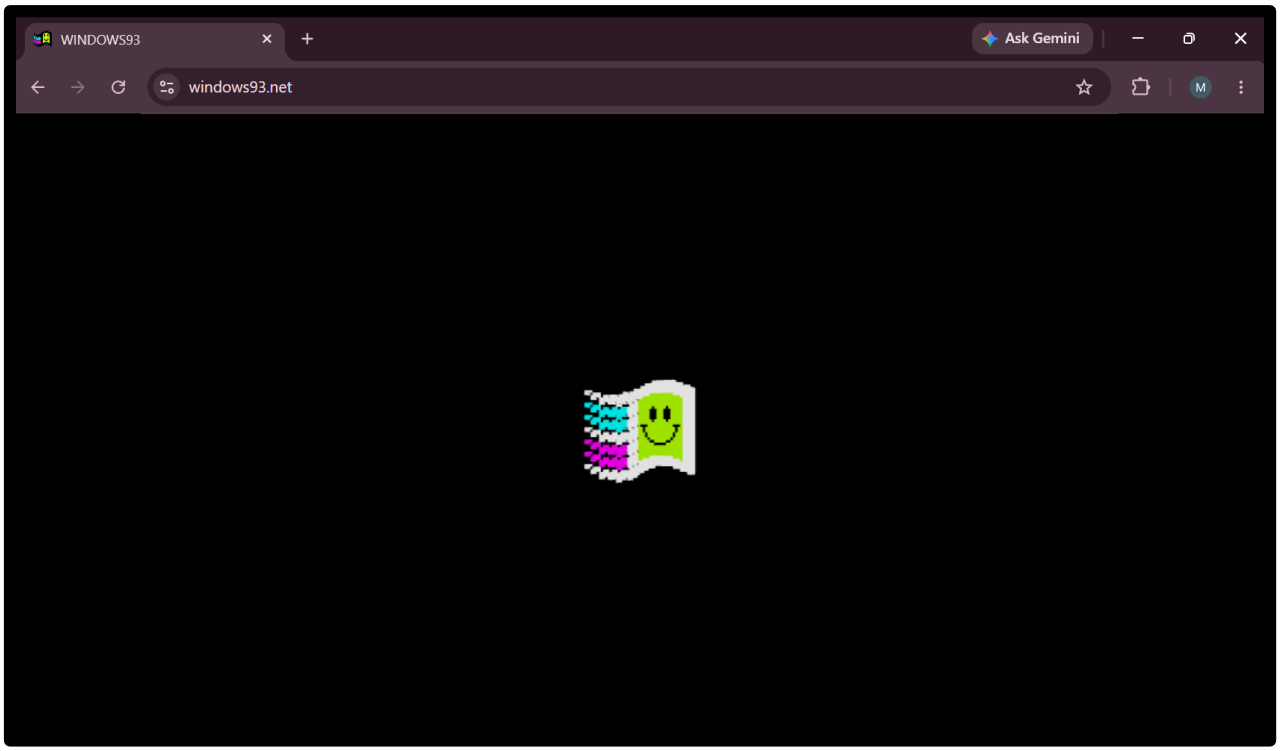


Windows 93:

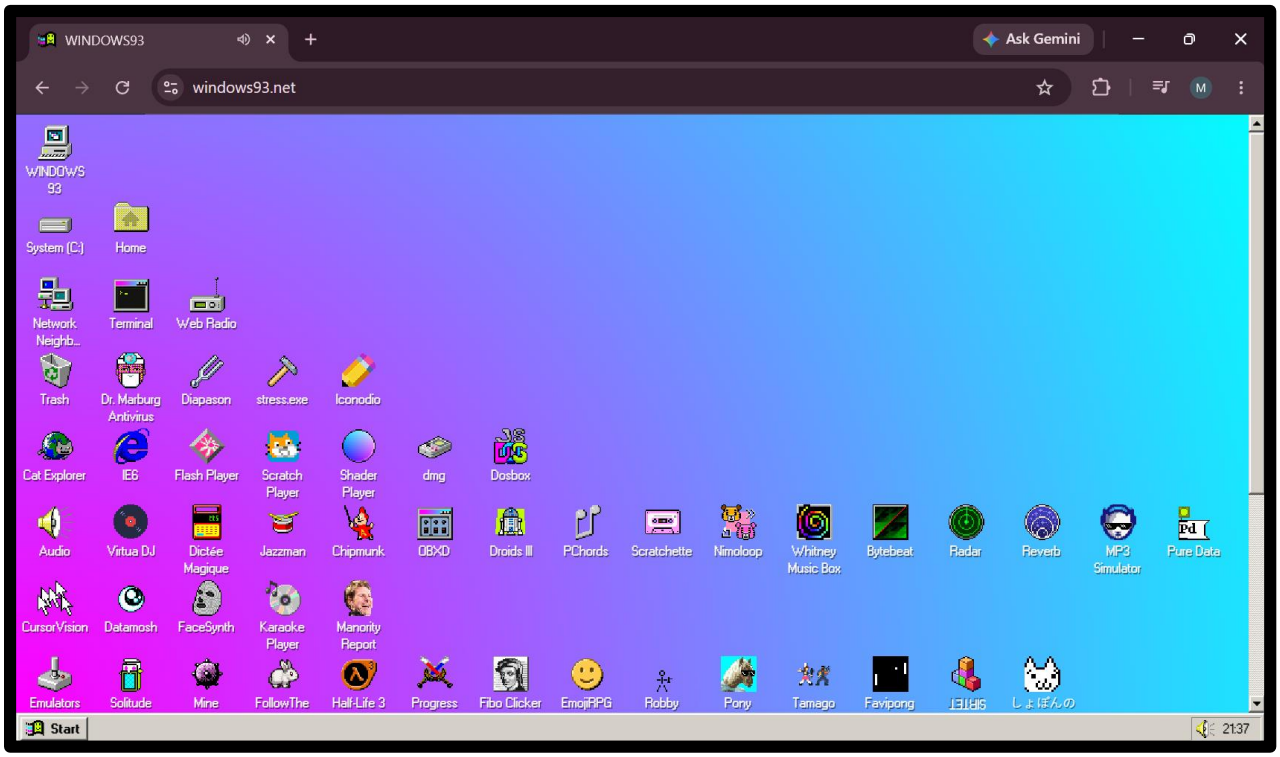
To see the interface of Windows 93, go online and write Windows 93 online.



Click on Windows 93

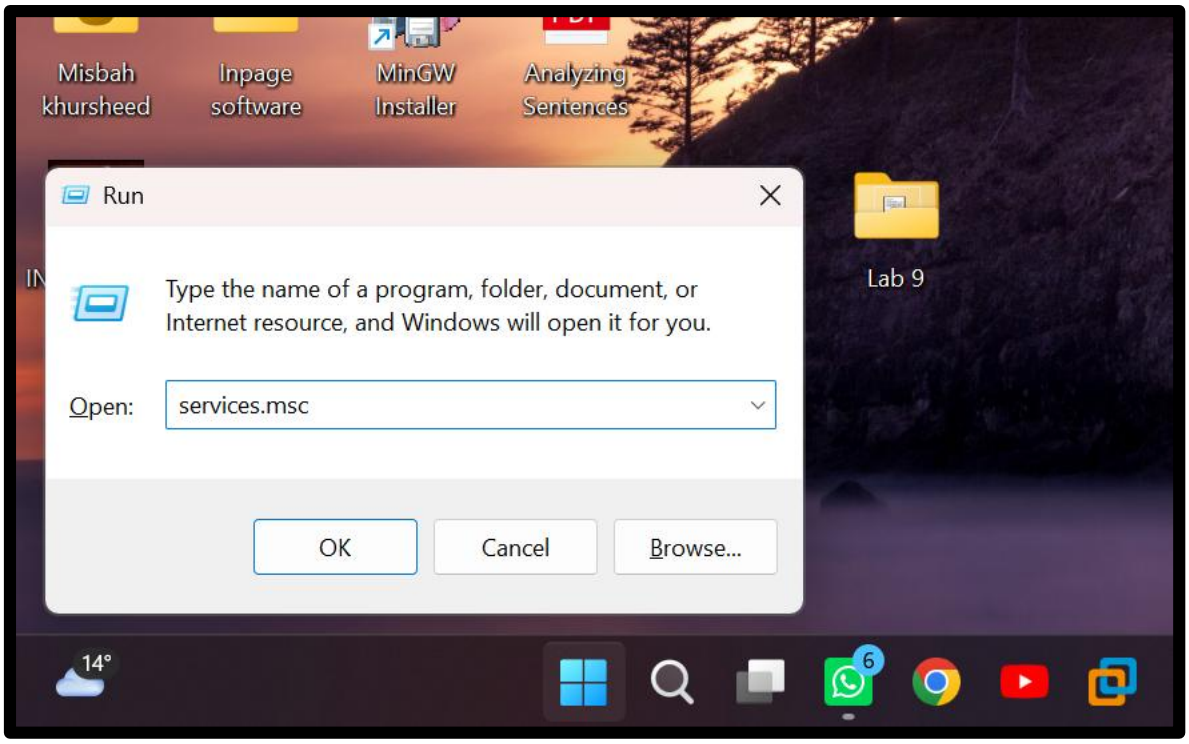


You can click on icons and see how windows 93 was working



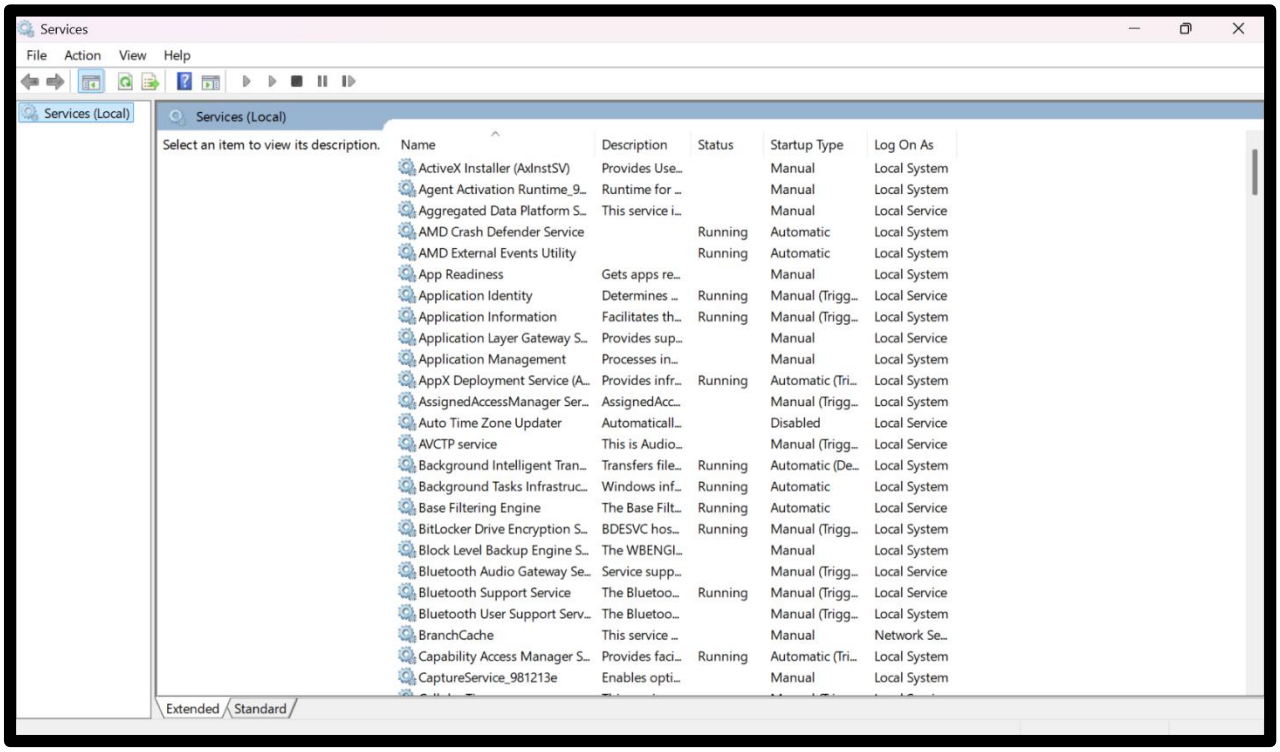
Operating systems services:

To check the services of Operating systems open the ‘run’ then, type “**services.msc**” and hit Enter or press OK.



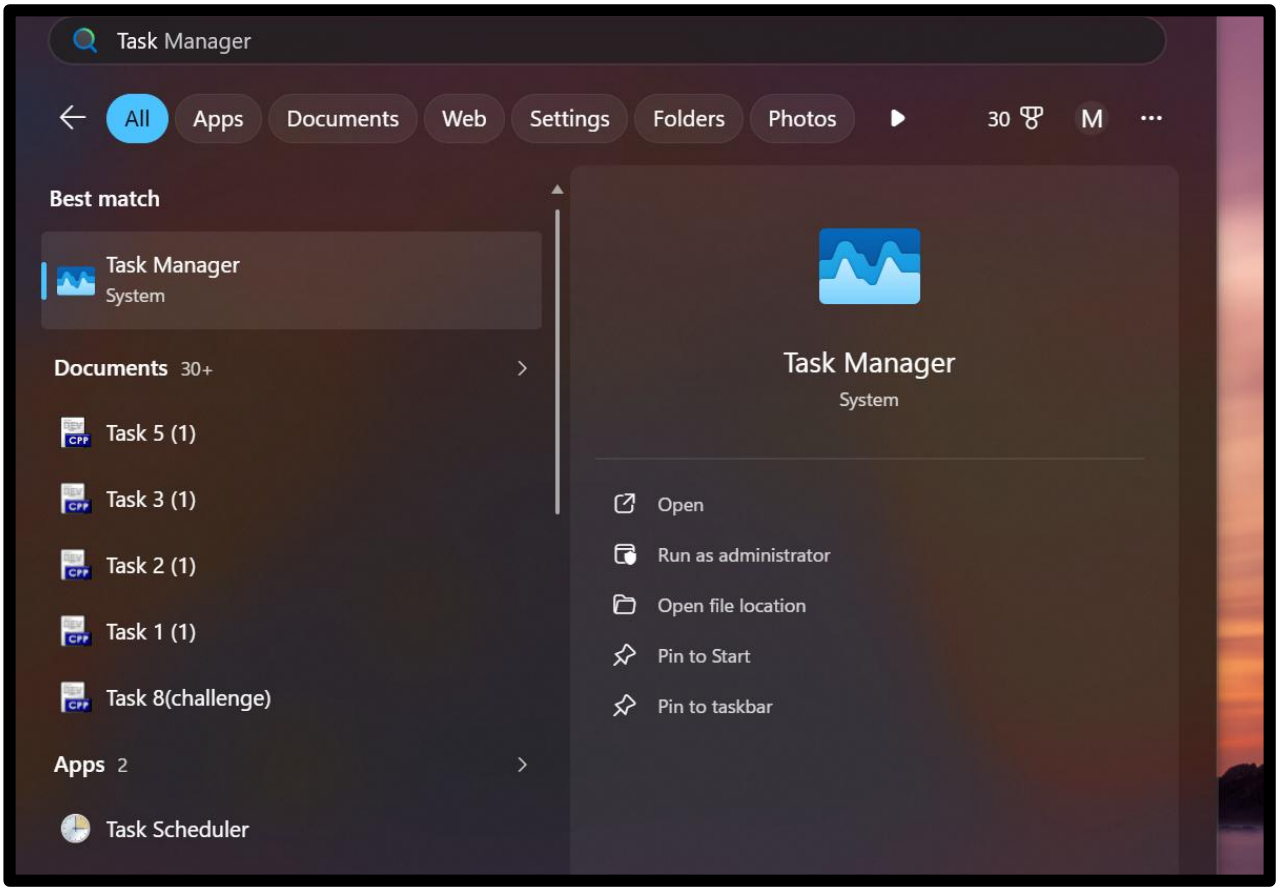
It's the Microsoft Management Console (MMC) file that loads the Services snap-in. It's used to administer Windows services. It provides a GUI to do things like: Start / stop / disable services. It shows you the services that operating systems performed.

Windows Services are applications that typically start when the computer is booted and run quietly in the background until it is shut down.



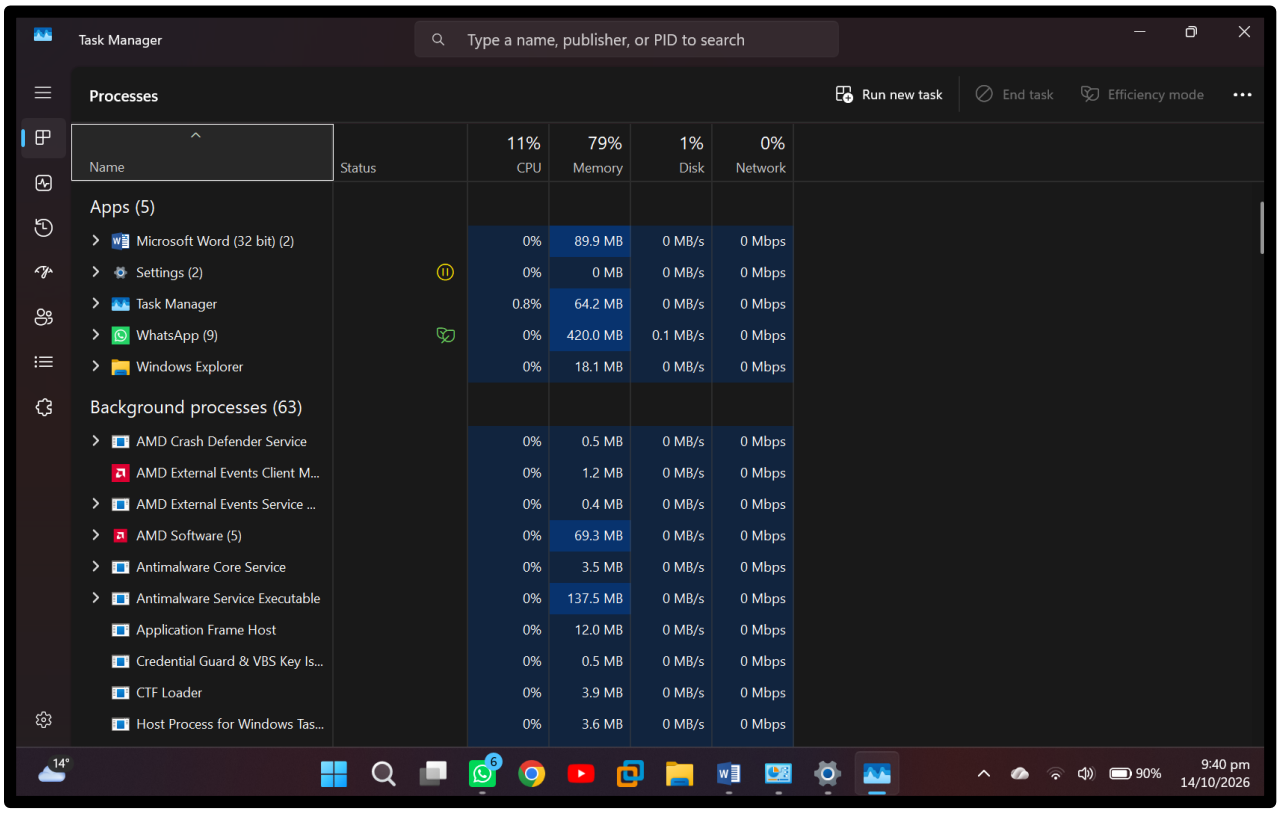
Task Manager:

The Windows Task Manager is a powerful tool packed with useful information, from your system's overall resource usage to detailed statistics about each process.

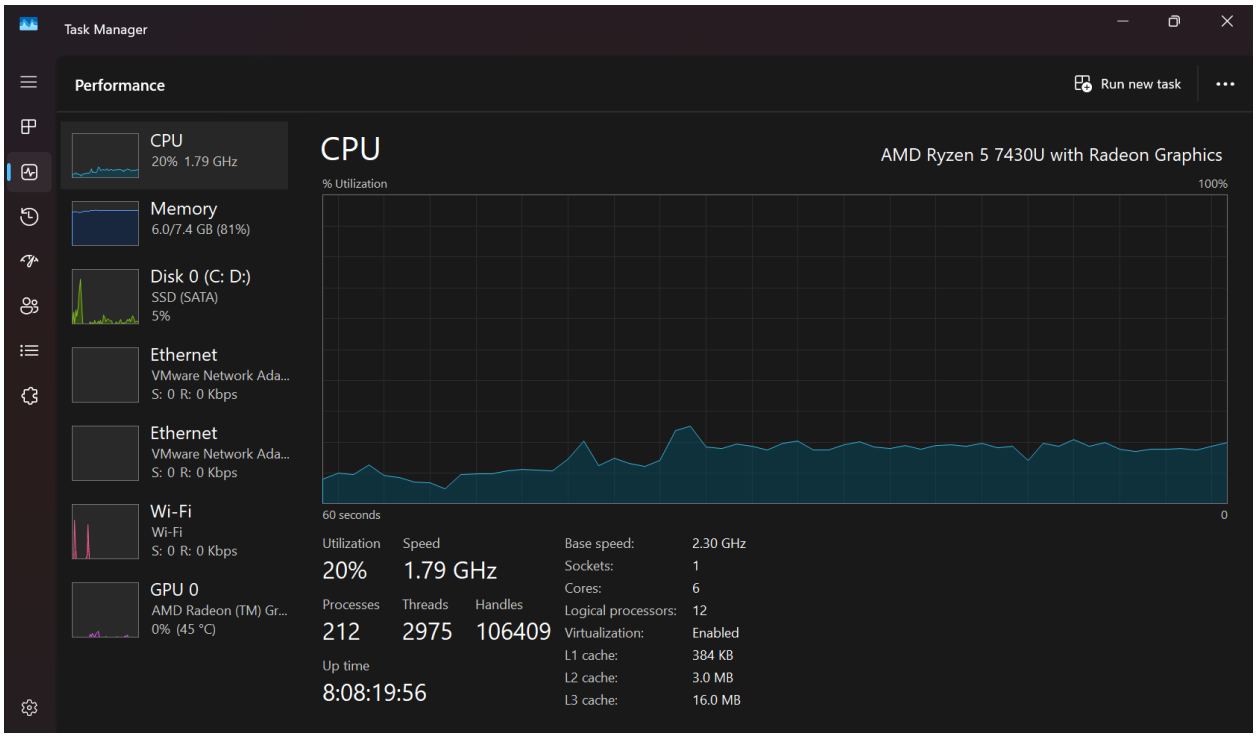


This shows you how much CPU (central processing unit) resources are currently in use on your system, and you can mouse over it to see memory, disk, and network usage

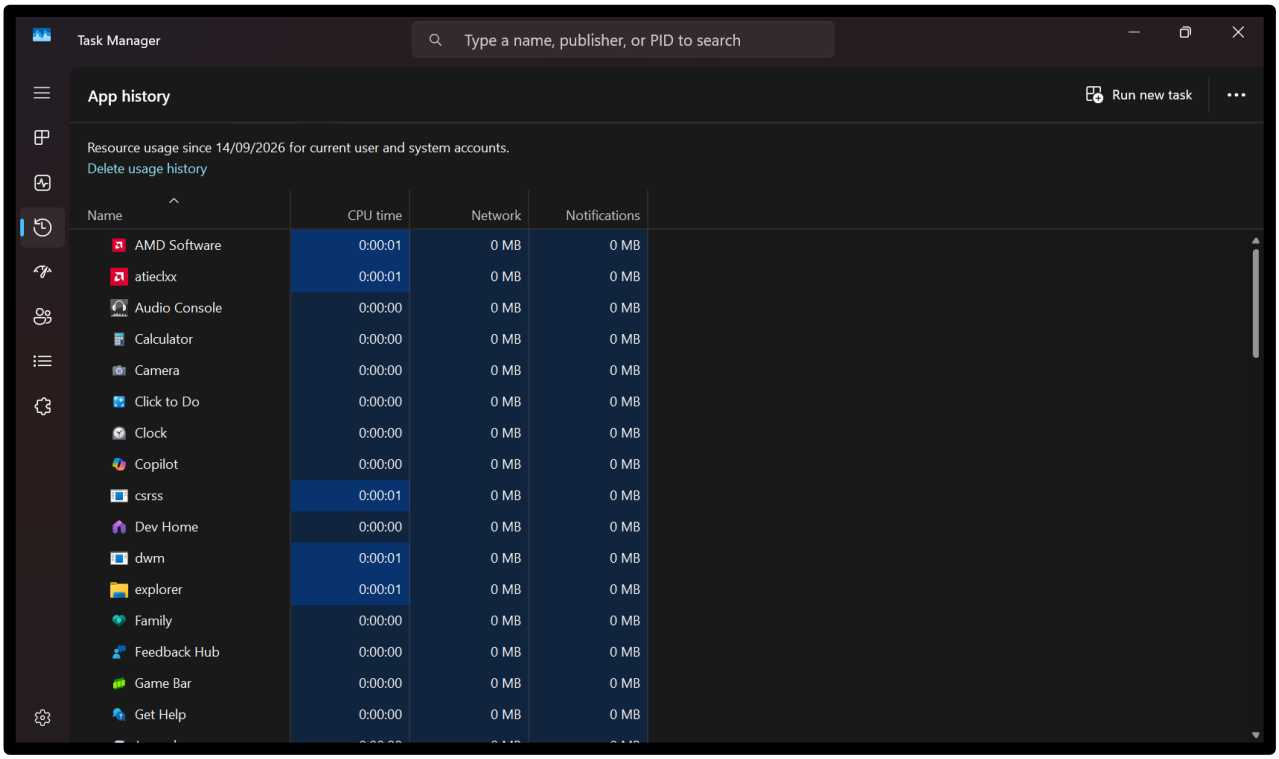
Processes shows a list of running applications and background processes on your system along with CPU, memory, disk, network, GPU, and other resource usage information.



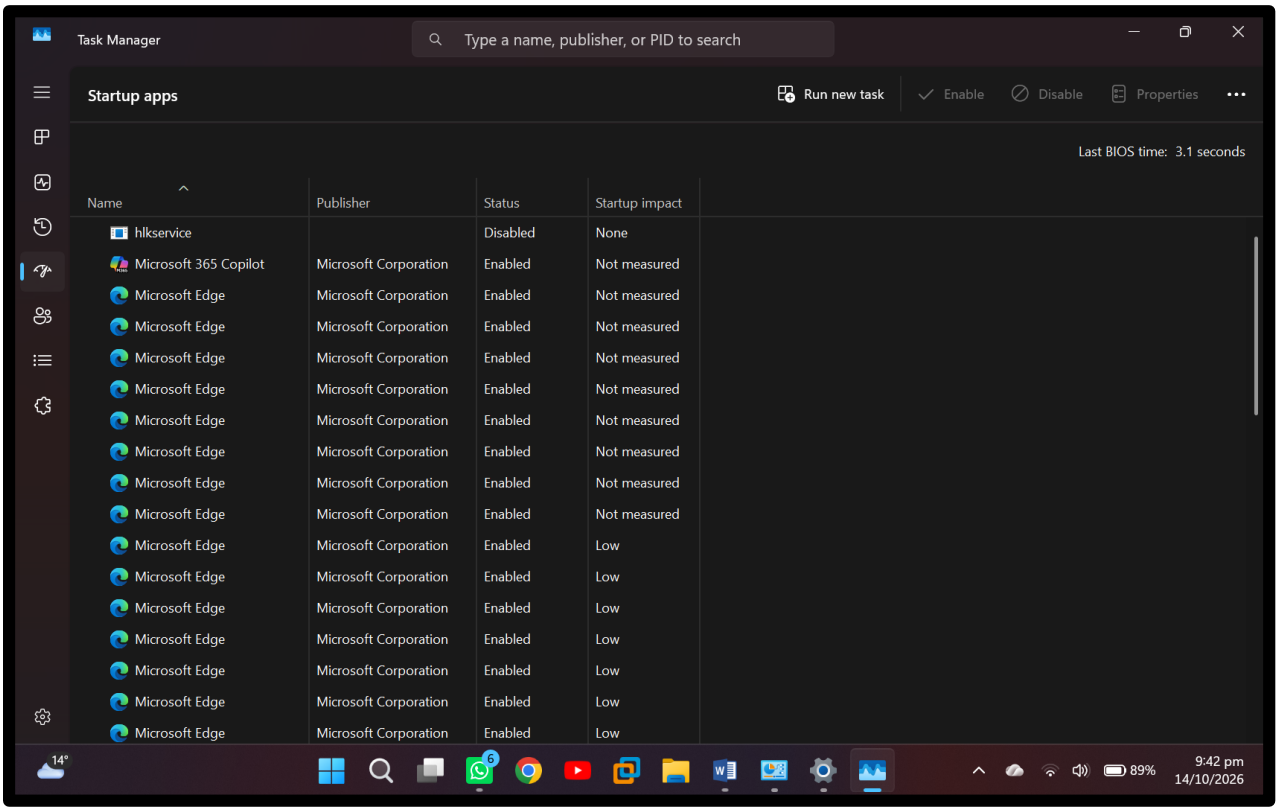
Performance shows Real-time graphs showing total CPU, memory, disk, network, and GPU resource usage for your system. You'll find many other details here, too, from your computer's IP to the model names of your computer's CPU and GPU.



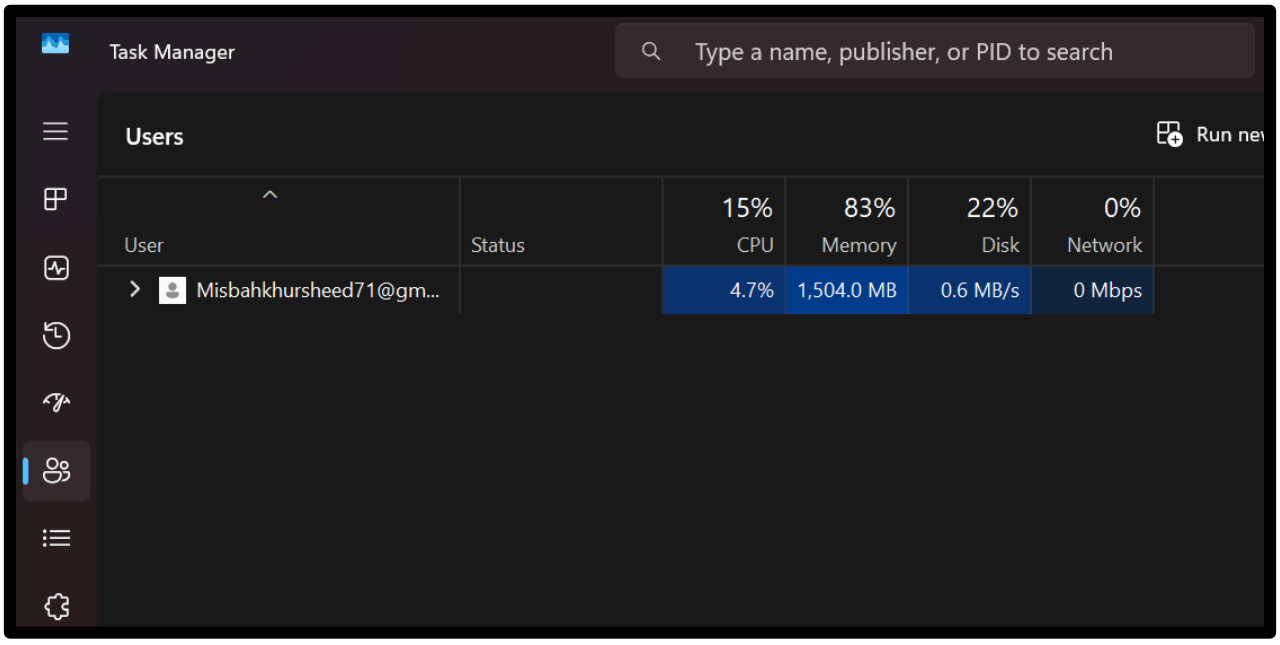
App History shows Information about how much CPU and network resources apps have used for your current user account.



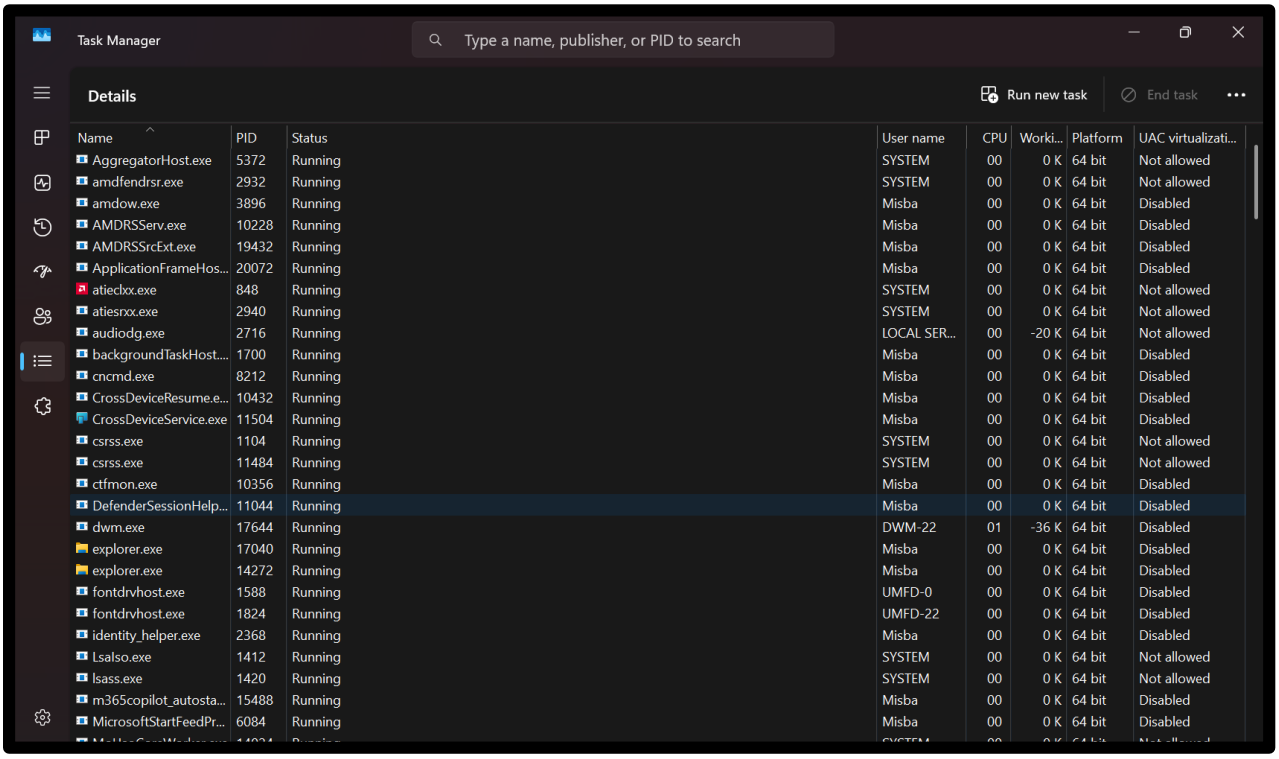
Startup shows a list of your startup programs, which are the applications Windows automatically starts when you sign into your user account. You can disable startup programs from here, although you can also do that from Settings > Apps > Startup.



Users shows the user accounts currently signed into your PC, how much resources they're using, and what applications they're running.

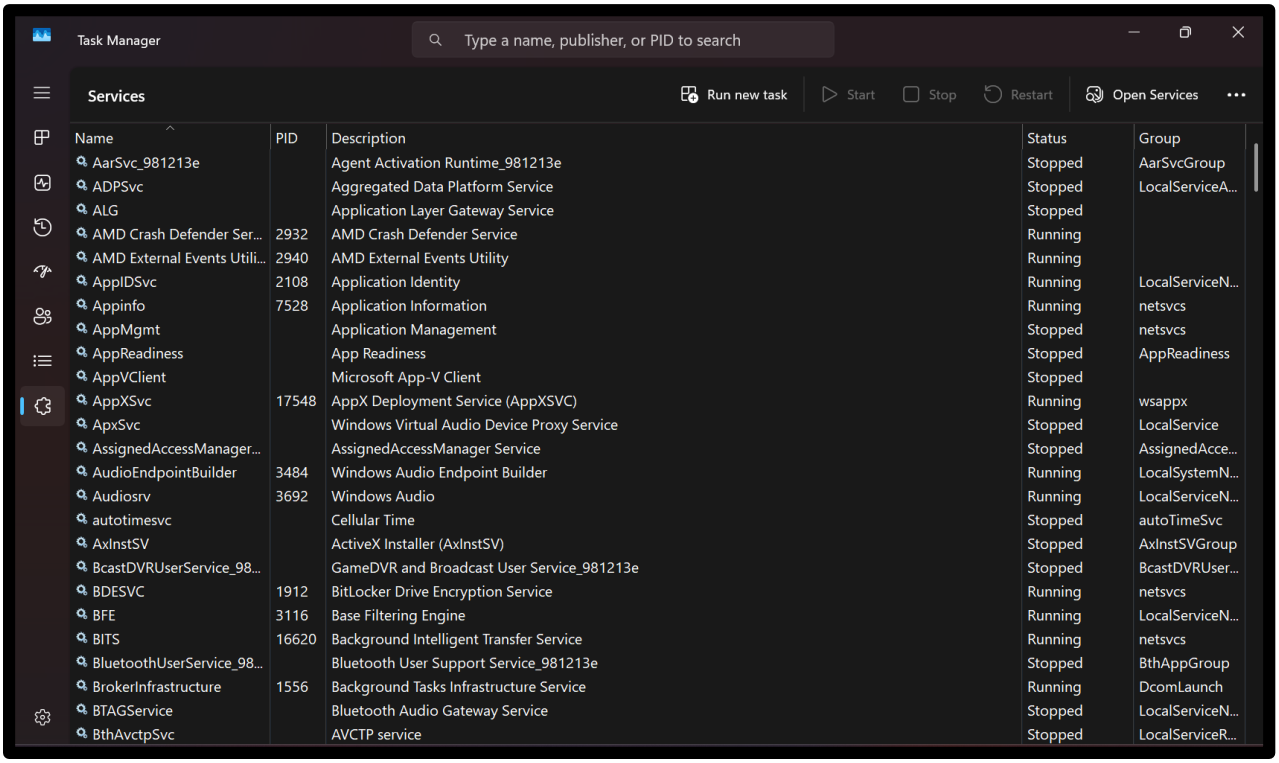


Details shows more detailed information about the processes running on your system. This is basically the traditional “Processes” tab from the Task Manager on Windows 11.



Name	PID	Status	User name	CPU	Worki...	Platform	UAC virtualizati...
AggregatorHost.exe	5372	Running	SYSTEM	00	0 K	64 bit	Not allowed
amdfendrsr.exe	2932	Running	SYSTEM	00	0 K	64 bit	Not allowed
amdow.exe	3896	Running	Misba	00	0 K	64 bit	Disabled
AMDRSServ.exe	10228	Running	Misba	00	0 K	64 bit	Disabled
AMDRSSrcExt.exe	19432	Running	Misba	00	0 K	64 bit	Disabled
ApplicationFrameHos...	20072	Running	Misba	00	0 K	64 bit	Disabled
atieclxx.exe	848	Running	SYSTEM	00	0 K	64 bit	Not allowed
atiesrxx.exe	2940	Running	SYSTEM	00	0 K	64 bit	Not allowed
audiodg.exe	2716	Running	LOCAL SER...	00	-20 K	64 bit	Not allowed
backgroundTaskHost....	1700	Running	Misba	00	0 K	64 bit	Disabled
cncmd.exe	8212	Running	Misba	00	0 K	64 bit	Disabled
CrossDeviceResume.e...	10432	Running	Misba	00	0 K	64 bit	Disabled
CrossDeviceService.exe	11504	Running	Misba	00	0 K	64 bit	Disabled
csrss.exe	1104	Running	SYSTEM	00	0 K	64 bit	Not allowed
csrss.exe	11484	Running	SYSTEM	00	0 K	64 bit	Not allowed
ctfmon.exe	10356	Running	Misba	00	0 K	64 bit	Disabled
DefenderSessionHelp...	11044	Running	Misba	00	0 K	64 bit	Disabled
dwm.exe	17644	Running	DWM-22	01	-36 K	64 bit	Disabled
explorer.exe	17040	Running	Misba	00	0 K	64 bit	Disabled
explorer.exe	14272	Running	Misba	00	0 K	64 bit	Disabled
fontdrvhost.exe	1588	Running	UMFD-0	00	0 K	64 bit	Disabled
fontdrvhost.exe	1824	Running	UMFD-22	00	0 K	64 bit	Disabled
identity_helper.exe	2368	Running	Misba	00	0 K	64 bit	Disabled
LsaIso.exe	1412	Running	SYSTEM	00	0 K	64 bit	Not allowed
lsass.exe	1420	Running	SYSTEM	00	0 K	64 bit	Not allowed
m365copilot_autosta...	15488	Running	Misba	00	0 K	64 bit	Disabled
MicrosoftStartFeedPr...	6084	Running	Misba	00	0 K	64 bit	Disabled

Services shows information about the services running on your system. It provides details about each service including its name, description, status, and group.



Name	PID	Description	Status	Group
AarSvc_981213e		Agent Activation Runtime_981213e	Stopped	AarSvcGroup
ADPSvc		Aggregated Data Platform Service	Stopped	LocalServiceA...
ALG		Application Layer Gateway Service	Stopped	
AMD Crash Defender Ser...	2932	AMD Crash Defender Service	Running	
AMD External Events Utili...	2940	AMD External Events Utility	Running	
AppIDSvc	2108	Application Identity	Running	LocalServiceN...
Appinfo	7528	Application Information	Running	netsvcs
AppMgmt		Application Management	Stopped	netsvcs
AppReadiness		App Readiness	Stopped	AppReadiness
AppVClient		Microsoft App-V Client	Stopped	
AppXSvc	17548	AppX Deployment Service (AppXSVC)	Running	wsappx
ApxSvc		Windows Virtual Audio Device Proxy Service	Stopped	LocalService
AssignedAccessManager...		AssignedAccessManager Service	Stopped	AssignedAcce...
AudioEndpointBuilder	3484	Windows Audio Endpoint Builder	Running	LocalSystemN...
Audiosrv	3692	Windows Audio	Running	LocalServiceN...
autotimesvc		Cellular Time	Stopped	autoTimeSvc
AxInstSV		ActiveX Installer (AxInstSV)	Stopped	AxInstSVGroup
BcastDVRUserService_98...		GameDVR and Broadcast User Service_981213e	Stopped	BcastDVRUser...
BDESVC	1912	BitLocker Drive Encryption Service	Running	netsvcs
BFE	3116	Base Filtering Engine	Running	LocalServiceN...
BITS	16620	Background Intelligent Transfer Service	Running	netsvcs
BluetoothUserService_98...		Bluetooth User Support Service_981213e	Stopped	BthAppGroup
BrokerInfrastructure	1556	Background Tasks Infrastructure Service	Running	DcomLaunch
BTAGService		Bluetooth Audio Gateway Service	Stopped	LocalServiceN...
BthAvctpSvc		AVCTP service	Stopped	LocalServiceR...